# Dr. Ravi Kumar Bandapelli

## Experimental Condensed Matter Physics Research Scientist | Spintronics | 2D Materials | Thin-Film Magnetism

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### RESEARCH PROFILE

Experimental quantum materials researcher specializing in spintronics, nanoscale device physics, and low-dimensional systems. Extensive experience in nanofabrication, thin-film growth, and low-temperature transport measurements. Experienced in SQUID and PPMS-based characterization (Ph.D. and IISc post doc), and currently working on magneto-transport using a home-built cryogenic system. Seeking to expand knowledge into magnonics and optical control over the magnetization beside the electrical control.

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### RESEARCH INTERESTS

Quantum materials • Strongly correlated systems • Spintronics
Topological materials • Superconductivity • Magnetotransport
2D van der Waals heterostructures • Thin-film magnetism

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### PROFESSIONAL EXPERIENCE

**Postdoctoral Researcher**

**Carnegie Mellon University, Pittsburgh, USA**
*2023 – Present*

- Fabrication of van der Waals heterostructure devices using several quantum materials like graphene, hBN, $Cr_2Ge_2Te_6$, $Fe_3GeTe_2$, $WTe_2$, $TaIrTe_4$, $Fe_3GaTe_2$, etc.
- Nanofabrication using e-beam lithography, optical lithography, and dry-transfer stacking
- Thin-film deposition via UHV sputtering and e-beam evaporation
- Low-temperature magnetotransport measurements using a **home-built cryogenic system**
- Developed nanoscale devices enabling observation of unidirectional magnetoresistance and in-plane anomalous Hall effect
- Collaborated on spectroscopic studies (ARPES, XMCD, TR-MOKE)
- Research direction includes extending device studies to exfoliated systems derived from high-quality bulk crystals

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**Institute of Eminence Postdoctoral Fellow**

**Indian Institute of Science (IISc), Bangalore, India**
*2022 – 2023*

- Investigated interfacial Dzyaloshinskii–Moriya interaction in CoFeB/Pt heterostructures
- Magnetic characterization using **SQUID, VSM, MOKE, and Kerr microscopy**
- Studied interfacial magnetism, anisotropy, and spin–orbit effects

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**Dr. D. S. Kothari Postdoctoral Fellow**

**Indian Institute of Science (IISc), Bangalore, India**
*2018 – 2021*

- Demonstrated current-induced magnetization switching in perpendicular magnetic anisotropy systems
- Investigated spin–orbit torque phenomena in magnetic heterostructures
- Performed nanofabrication and MOKE measurements
- Experience with **SQUID magnetometry and PPMS-based measurements**

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**Research Associate**

**National Institute of Science Education and Research (NISER), India**
*2018*

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## EDUCATION

**Ph.D. in Experimental Condensed Matter Physics**
*School of Physics, University of Hyderabad, Hyderabad, Telangana-500 046, India* (2018)
*Supervisors:* **Prof. S. N. Kaul and Prof. S. Srinath**

- Thesis title: *Effect of spatial dimensionality on magnetic properties of* $Cr_{100-x}Fe_x$ *thin films above the critical concentration for ferromagnetism*
- Magnetic characterization using **SQUID and PPMS systems**

**M.Sc. in Condensed Matter Physics**
Osmania University, India (2009)

**B.Sc. in Mathematics, Physics, Chemistry**
Kakatiya University, India (2007)

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## SELECTED RESEARCH PROJECTS

**Van der Waals Spintronic Devices for Magnetotransport Studies**

*Carnegie Mellon University (2023–Present)*

- Fabricated heterostructures and devices using several 2D-quantum materials
- Designed nanoscale devices via e-beam lithography, reactive ion etching (RIE), and dry-transfer stacking
- Performed magnetotransport measurements using a **home-built cryostat**
- Observed **unidirectional magnetoresistance** and **in-plane anomalous Hall** effects
- Supported collaborative spectroscopic studies (ARPES, XMCD, TR-MOKE)

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**Spin–Orbit Torque and Magnetization Switching in CoFeB/Pt Systems**

*IISc Bangalore (2018–2023)*

- Studied current-induced field-free magnetization switching in PMA heterostructures
- Investigated interfacial Dzyaloshinskii–Moriya interaction and anisotropy engineering
- Performed measurements using **SQUID, VSM, MOKE, and PPMS**

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**Magnetic Phase Transitions in Cr–Fe Thin Films**

*Ph.D. in Physics*
*Thesis title: "Effect of spatial dimensionality on magnetic properties of $Cr_{100-x}Fe_x$ thin films above the critical concentration for ferromagnetism"*

- Investigated finite-size effects and dimensionality effects in magnetic systems
- Studied magnetic transitions using **SQUID and PPMS measurements**

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## TECHNICAL EXPERTISE

**Low-Temperature & Magnetic Measurements**

- SQUID magnetometry *(Ph.D., IISc)*
- PPMS (magnetic measurements) *(Ph.D.)*
- Home-built cryogenic transport systems *(CMU)*
- VSM, MOKE *(IISc)*

**Materials & Fabrication**

- Thin-film deposition (UHV sputtering, e-beam evaporation)
- Nanofabrication (e-beam lithography, optical lithography, RIE, ion-milling)
- 2D material exfoliation and dry-transfer stacking

**Characterization**

- AFM, SEM, Raman spectroscopy, X-ray reflectivity (XRR)

**Additional**

- Experience with quantum materials and heterostructures
- Strong interest in **optical control over magnetization**

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## PUBLICATIONS

**Research publications (First author)**

1. I-Hsuan Kao†, <u>**B. Ravi Kumar†**</u>, et al., “In-plane anomalous Hall effect in a low-dimensional system”, *Nature Materials* (2026) *(†equally contributed)*
2. <u>**B. Ravi Kumar**</u>, et al., “Tuning the interfacial Dzyaloshinskii–Moriya interaction in perpendicularly magnetized CoFeB system”, *Journal of Physics D: Applied Physics* **55**, 445004 (2022).
3. <u>**B. Ravi Kumar**</u>, et al., “Enhancing the properties of CdO thin films by co-doping with Mn and Fe for photodetector applications”, *Sensors and actuators* **219,** 112544 (2021).
4. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Anomalous softening of magnon modes in the reentrant state in $Cr_{70}Fe_{30}$ thin films”, *J. Magn. Magn. Mater*. **469**, 681 (2019).
5. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Thickness as a control parameter for magnetocaloric effect in $Cr_{75-x}Fe_{25+x}$ (x = 0, 5) thin films”, *J. Magn. Magn. Mater.* **460**, 312 (2018).
6. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Isotropic-Heisenberg to isotropic-dipolar crossover and finite-size scaling in $Cr_{75-x}Fe_{25+x}$ (x = 0, 5) thin films”, *J. Mag. Mag. Mater.* **448**, 3 (2018).
7. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Nature of reentrant state in $Cr_{75-x}Fe_{25+x}$ (x = 0, 5) thin films”, *J. Magn. Magn. Mater*. **401**, 539 (2016).
8. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Magnetic order-disorder phase transition in $Cr_{70}Fe_{30}$ thin films”, *J. Alloys Compd*. 652, 479 (2015).
9. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Irreversibility lines in the T -H phase diagram of $Cr_{70}Fe_{30}$ thin films”, *Physica B* 448, 140 (2014).

**Research publications (Co-author)**

1. Vanga Ganesh, <u>**B. Ravi Kumar**</u>, et al., “Electrocatalytic Degradation of Rhodamine B Using Li-Doped ZnO Nanoparticles: Novel Approach”, Materials **16**, 1177 (2023).
2. Harish, <u>**B. Ravi Kumar**</u>, et al., “Structural, electrical, and optical properties of rare-earth Sm3+ doped SnO2 transparent conducting oxide thin films for optoelectronic device applications: Synthesized by the spin coating method”, Optical materials **133**, 112993 (2022).
3. Pittala Suresh, <u>**B. Ravi Kumar**</u>, et al., “Lattice effects on the multiferroic characteristics of (La, Ho) co-substituted $BiFeO_3$”, *J. Alloys Compd*. **863**, 158719 (2021).
4. V. Ganesh, <u>**B. Ravi Kumar**</u>, Yugandhar Bitla, I. S. Yahia, S. AlFaify, “Structural, Optical and Dielectric Properties of Nd Doped NiO Thin Films Deposited with a Spray Pyrolysis Method”, J. Inorg. Organomet. Polym. Mater. **31**, 2691 (2021).

**Publications in conference proceedings**

1. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Thickness induced crossover from the ferromagnetic to cluster spin glass state in $Cr_{70}Fe_{30}$ thin films”, *AIP Conf. Proc.* **1665**, 030032 (2015).
2. <u>**B. Ravi Kumar**</u> and S. N. Kaul, “Critical behavior near ferromagnetic-paramagnetic phase transition in ion beam sputtered $Cr_{70}Fe_{30}$ thin films”, *AIP Conf. Proc.* **1536**, 953 (2013).

**Manuscripts under review**

1. Raghvendra, **B. Ravi Kumar**, et al., *[Enabling Electrical Readout of Néel vector reversal in a van der Waals Antiferromagnet]* Submitted to **Nature Communications** *(under review)* (reference number: NCOMMS-26-049950-T) arxiv: https://doi.org/10.48550/arXiv.2606.24527
2. Sandy, Aalok Tiwari,…. **B. Ravi Kumar**, Jyoti Katoch, Simranjeet Singh, et al., *[Visualization of Tunable Electronic Structure of Monolayer $TaIrTe_4$]* Submitted to **Nature Communications** (reference number: NCOMMS-26-004797) *(under review)*
3. Wenyi, **B. Ravi Kumar**, Roland Kawakami, et al., *[Ultrafast light-induced magnetoelectric effect in van der Waals magnetic semiconductor heterostructures]* Submitted to **Physical Review Letters** *(under review) arxiv:* https://doi.org/10.48550/arXiv.2604.19080

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**Ongoing works:**

1. **B. Ravi Kumar†**, I-Hsuan Kao†, et al., *[Tilted Spin Polarization Enabled Multidimensional Unidirectional Magnetoresistance]* (ready for submission, †*equally contributed*)
2. Aalok Tiwari†,.., **B. Ravi Kumar,** et al., *[Spatially Resolved Magnetism and Twist-Induced Domain Reconfiguration in Atomically Thin CrSBr]* (ready for submission)
3. Zhenhong Cui, **B. Ravi Kumar,** *[Gate tunability of ferroelectricity in $WTe_2$/CGT bilayer device]* (manuscript in preparation)
4. Aalok Tiwari, …., **B. Ravi Kumar,** et al., *[Angle resolved Photoemission studies of microelectronic Devices and Quantum structures]* (manuscript in preparation)
5. **B. Ravi Kumar** and P. S. Anil Kumar [*Tuning the field-like torque using multilayer stacks in CoFeB/Pt heterostructures*] (manuscript in preparation)

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**INVITED TALKS & PRESENTATIONS**

- Oral talk on results of Unidirectional magnetoresistance and Anomalous Hall effects in APS march meeting 2024 and 2025 and MMM & Intermag conference 2024
- Invited talk: Symposium on Magnetism and Spintronics (2021)
- International Conference on Magnetism and Magnetic Materials
- DAE Solid State Physics Symposium
- Frontiers in Physics Conference
- Best poster awards

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**FELLOWSHIPS & AWARDS**

- Dr. D. S. Kothari Postdoctoral Fellowship (IISc, India)
- Institute of Eminence Postdoctoral Fellowship (IISc, India)

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**REFERENCES**

**Prof. Simranjeet Singh**
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Relationship: Ph.D. Advisor
I will request them to arrange the reference letters upon your interest